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LAUDS FACTORIES

Open Call 2 | LAUDS Replication

Production Process Blueprint

DATA-RIDE

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Project coding LAUDSREP-01-19	*DATA-RIDE: Data and Advocacy for Transforming Active Riding & Infrastructure for a Democratic Environment*	
Deliverable title Production Process Blueprint: Data-Ride		**Due date** 01.10.2026
Hybrid Team entities **Lead partner:** re:edu GmbH & Co. KG **Partner 1: Laura Sobral** **Partner 2: Global Innovation Gathering (GIG)**		
Challenge Category Sustainable urban mobility	**Challenge Title** Citizen-led sensing and data-driven advocacy for safer cycling infrastructure	**Host LAUDS Factory** Université de Lorraine

1. Product and process Description

Please provide detailed information on the product or solution you have developed in your project for the LAUDS Open Call 2, including the elements detailed below.

Production Process

The technical development included adding a second Time-of-Flight (ToF) sensor to the senseBox setup for implementation during the Nancy campaign. Measuring lateral distances on both sides enabled exploration of the space available (corridors) around cyclists, complementing the analysis of potential overtaking events.

The sensing setup comprised a microcontroller (ESP32-S2), two ToF distance sensors, a particulate-matter sensor (SPS30), temperature and humidity sensors (HDC1080), an inertial measurement unit for motion and road-surface indicators (MPU6050), and a GPS data from the smartphone. Devices were mounted on participants' bicycles and connected to smartphones for recording and uploading data. Physical materials included sensor enclosures, mounting components, and a mounting prototype developed with local laboratory support. Online communication materials (Poster and Policy Recommendation) translated the findings into accessible visuals for public discussion.

The analysis used GPS position, speed, left- and right-side distances, particulate-matter measurements, and motion-based indicators of stopping and road-surface conditions. The processed dataset contained 193,328 observations across 357 trips, covering more than 1,000

kilometres and approximately 84 hours of valid recorded movement. Recordings were linked to OpenStreetMap street segments and combined with local datasets on cycling infrastructure, parking, and tree presence.

These inputs supported the production of a series of maps visualising the recorded cycling routes and different dimensions of the cycling experience. The maps included the spatial distribution of potential close-overtaking events, available lateral corridor widths during rides, road-surface roughness, and particulate-matter measurements. Additional spatial analyses combined the sensor recordings with cycling infrastructure and on-street parking data, making it possible to explore where constrained cycling space and potential close passes occurred in relation to different street and infrastructure conditions.

For the overtaking maps, potential events were identified by combining short-term patterns in left-side ToF distance measurements with the on-device machine-learning overtaking indicator. The analysis used two-second windows, filtered invalid distance measurements and stationary periods, removed repeated patterns potentially associated with parked cars, and merged nearby detections into individual events. These were mapped as potential overtaking events, rather than confirmed overtakes. Further local calibration and video-based validation would be needed to improve the reliability of the detection method

The local ecosystem involved the Métropole du Grand Nancy, the Lorraine Smart Cities Living Lab, EDEN, Atelier Dynamo,, Laboratoire Sauvage, volunteer cyclists, and workshop participants. Contributions ranged from hosting workshops and presentations to organising collections and supporting local outreach.

Resources included sensing devices, bicycles, smartphones, computing equipment, fabrication facilities, and digital services. Energy was required for charging, fabrication, data processing, and online services, although consumption was not quantified during the project.

Product Maturity

The project began with an existing senseBox:bike platform and a proposal to extend its functionality and apply it through a participatory campaign in Nancy. The support of LAUDS Factories was particularly important in enabling the team to move beyond technical development and test the system in a real-world setting with local cyclists.

By the end of the project, the system had been deployed with ten volunteers and supported a processed dataset, spatial analyses, public discussions, communication outputs and policy recommendations. Open sessions enabled participants to discuss sensor observations alongside their cycling experiences and identify practical challenges in using the technology.

Maturity differed between components. The following initial and final TRL values are provisional retrospective self-assessments, subject to confirmation against technical records.

Component	Initial TRL	Final TRL	Supporting rationale
senseBox:bike	4	6	The platform was adapted and deployed with volunteers in Nancy. Changes included firmware modifications, adaptations to cloud data transmission and French-language support.
Second distance sensor and adapted firmware	2	6	The bilateral configuration was implemented and demonstrated through field data collection. Measurement accuracy requires further controlled testing and validation across different operating conditions.
Acrylic mounting support	2	5	The support was fabricated at the Fab Lab Nancy and tested in May. Longer-duration testing is needed to assess durability and retention across different bicycle configurations.
Data processing and spatial mapping workflow	2	6	Python scripts and QGIS processing were applied to campaign data to produce analytical outputs and maps. Some spatial parameters still require documentation for full reproducibility.
Overtaking classification	3	3	Potential events were identified, but reliable local calibration and synchronised video validation remain incomplete. A validation protocol is needed to establish detection performance.
Safe Passing Corridor Index	2	3	Bilateral distance indicators were calculated and mapped as an exploratory proof of concept. Some data validation and a expand data collection campaing still needed.
Policy Recommendation Paper	N/A	N/A	Connected exploratory data findings with local discussions about cycling conditions, participants' experiences and priorities. It translated these contributions into recommendations for local advocacy and dialogue with public authorities. Policy uptake remains to be assessed.

Visual Toolkit	N/A	N/A	Combined guidance on sensor technology with citizen-led methods of organisation, participation and advocacy. It provides a resource for adapting the approach to other places, linking data collection with collective discussion and action. Replication outside Nancy remains to be evaluated.

Process and Blueprint

The final blueprint organises DATA-RIDE across four areas:

Blueprint area	Main contribution
Core Product	Citizen-generated cycling data, shared interpretation, and Policy Recommendation materials.
Implementation & Production	Prototype development, supported collection, and open documentation.
Ecosystem Partners	Actor mapping, partner-based interpretation, and policy recommendations.
Community & Customers	Accessible volunteer participation, public communication, and advocacy resources.

Beyond the product itself, the project matured through the creation of a local partner network, direct involvement of cyclists as participants and co-interpreters, and stronger connections with public authorities through the presentation and discussion of results. The Nancy campaign also helped clarify where technical support, calibration, documentation, and facilitation are needed for future deployments.

The sustainability approach therefore goes beyond maintaining the open-source hardware. It points toward a service model around an open-source core, including campaign setup, sensor adaptation and calibration, technical support, data processing and interpretation, participatory workshops, and policy-oriented communication materials. This could support future collaboration with municipalities, research institutions, cycling organisations, or other local partners.

The blueprint also shows that replication requires more than copying the device. It requires a combination of open technology, technical support, local participation, contextual interpretation, partner coordination, and output such as the toolkit and policy recommendation materials.

Community ownership and an active partner network are therefore as important to long-term sustainability as the technology itself.

Documentation

Project documentation is available in the DATA-RIDE GitHub repository, including processing scripts, methodology, maps, the Visual Toolkit and communication materials.

Project-authored processing scripts are licensed under the MIT License. The DATA-RIDE Blueprint is licensed under CC BY 4.0. Project-authored maps, toolkit and communication materials are licensed under CC BY-SA 4.0, unless otherwise indicated. Third-party content retains its original licences and attribution requirements.

Source: https://github.com/reedu-reengineering-education/dataride-lauds

Firmware source:
https://github.com/sensebox/senseBox-bike-Arduino/tree/main/senseBox-bike-nancy

The public history of the senseBox-bike-nancy folder records the following changes:

Date	Firmware change	Commit
December 2025	Added a Nancy-specific version supporting a second Time-of-Flight distance sensor for bilateral measurements.	3a46e98
December 2025	Updated the SPS30 particulate-matter sensor library.	45f51f3
December 2025	Added battery-monitoring functionality.	5639716
December 2025	Adapted sensor communication through the I²C multiplexer and updated lateral-distance reading handling.	0a64eff
January 2026	Assigned a dedicated Bluetooth characteristic UUID to right-side distance measurements, replacing the previously reused temperature UUID.	70250f9
January 2026	Added support for the ICM20948 motion sensor.	fbdceca

French-language support was an application adaptation and is therefore reported separately from firmware changes.

2. Impact

Exploitation plan

DATA-RIDE's exploitation approach centres on the reuse of the tested technology and participatory methodology as an integrated service package. Potential applications include local cycling campaigns commissioned by municipalities or urban mobility departments, research collaborations, educational activities, and citizen-led initiatives supporting evidence-informed mobility discussions.

The service could combine:

- senseBox:bike devices and technical setup;
- local adaptation and an improved calibration and validation protocol;
- support for participatory data-collection campaigns;
- data processing, spatial analysis, maps and visualisations;
- local workshops to interpret the results together with cyclists and other stakeholders;
- communication and policy recommendation materials to support dialogue with local authorities.

Rather than offering the technology as a standalone product, the aim is therefore to provide a complete campaign and analysis service around an open-source technology.

The immediate priorities are to improve the validation protocol, maintain accessible and reusable documentation, and consolidate the technical, analytical and participatory components into a clearly defined service package that can be replicated and adapted to different local contexts.

Sustainability and Community Involvement

- Open sustainability: The project builds on open-source technology and aims to maintain openly accessible documentation, methodologies, data-processing workflows, maps. This allows the Nancy stakeholders and future communities and partners to adapt the approach without depending on a closed technological system.
- Local sustainability: The Nancy campaign involved local cyclists, cycling organisations, researchers, and local public actors throughout the process. Participants contributed not only by collecting data, but also by identifying relevant cycling issues and later interpreting the results based on their local knowledge and lived experience.
- Community involvement: Workshops created a feedback loop between participants and the project team. Feedback led to practical adaptations, including French-language

materials and an alternative mounting solution for different bicycle configurations. (designed by University of Lorraine and tested during the second workshop).The interpretation workshop also connected the collected evidence back to the issues raised by participants at the beginning of the campaign.

- Ecological sustainability: By supporting cycling advocacy and evidence-informed discussions about cycling infrastructure, DATA-RIDE contributes to broader efforts to strengthen active and low-carbon mobility.

Key exploitation

The local discussions and the workshop, where sensor data were compared with cyclists' perceptions and experiences, generated not only maps but also four local directions for further exploration by the Nancy cycling community:

1. Promote safe overtaking and mutual respect.
2. Support new cyclists through peer learning.
3. Improve the quality and continuity of cycling infrastructure.
4. Continue developing citizen-generated mobility data.

Key Performance Indicators (KPIs)

KPI	Technical Annex	Reported result	Measurement method / clarification
Participants engaged	Local cyclists and cycling organisations; numerical target not specified.	10 volunteers	Participant registration and device-distribution records. This counts sensing volunteers.
Data-collection period	Three-week community sensing campaigns.	February–May 2026	Sensor timestamps and campaign records. This is the overall collection calendar.
Recorded trips	Not specified.	357 trips	Trip identifiers in the processed dataset, constructed using gaps greater than five minutes between observations from the same device.

Distance recorded	Not specified.	1,234.74 km	Reported GPS-based total.
Cycling time recorded	Not specified.	84 hours	Reported sensor-recording duration.
Participatory workshops	Two co-creation workshops to develop policy recommendations; launch training and other engagement activities also planned.	3 workshops: Launch, local hypotheses and data interpretation/policy discussion	Activity records and photographs; the Nancy partner holds attendance lists for some sessions.
Public disseminatio n events	Public presentations and dissemination workshops;	Fête du Vélo and an online presentation at Université de Lorraine	Presentation materials and event records.
Types of spatial analysis produced	Visual report covering overtaking corridors, air quality, temperature and road-surface bumpiness; numerical target not specified.	9 map themes	Final maps cover routes, standing hotspots, road surface, speed, corridors near parking, overtaking distance, infrastructure categories, PM2.5 and PM2.5/tree relationships. Temperature was not collected following replacement of the temperature sensor with the second distance sensor.
Local adaptations implemented	Develop and validate a dual-side passing-distance module; numerical target for other adaptations not specified.	Dual-distance configuration, plus 2 local adaptations: French-language support and a	Firmware repository, application materials and mounting documentation. Functional deployment was demonstrated;

		prototype of an alternative bicycle mounting support (designed by University of Lorraine and tested during the second workshop)	French-language support is an application change.
Open outputs produced	Open hardware documentation, including schematics and an assembly guide; validated, anonymised datasets and maps aggregated by street segment; visual report; policy recommendation; advocacy toolkit with communication materials.	Dataset, methodology, maps, Policy Recommendation Paper, Toolkit, Blueprint and communication materials	Project files and repository. Street-segment aggregation was provided in maps rather than as a separate published dataset.

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